

TOWARDS AGRICULTURAL CIRCULARITY: BIOMASS-DERIVED BIODEGRADABLE MULCH SHEETS WITH POTENTIAL FOR PHOTOCATALYTIC SOIL REMEDIATION

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01 INTRODUCTION

Agriculture consumes millions of tons of plastic mulch annually, generating persistent waste after use



03 RESEARCH GAP AND CONCEPT

EXISTING NATURAL CONDITIONS



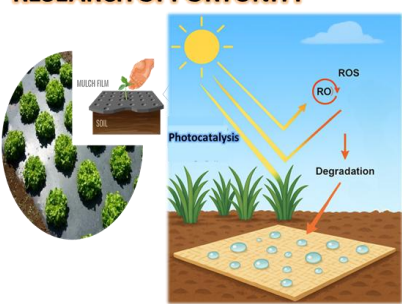
"Fundamental requirements of Photocatalysis"

CURRENT PROBLEMS



Our Concept: A biodegradable mulch capable of both physical soil protection and active pesticide degradation.

RESEARCH OPPORTUNITY



Gap & Opportunity

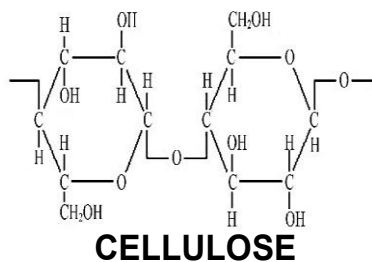
- Current mulches remain passive
- Photocatalysis rarely explored in agricultural mulch
- Visible-light remediation under field conditions underexplored

04 METHODOLOGY AND MATERIAL DESIGN

Agricultural Biomass as Feedstock



- ✓ Cellulose rich
- ✓ Easily available
- ✓ Low-cost
- ✓ Biodegradable



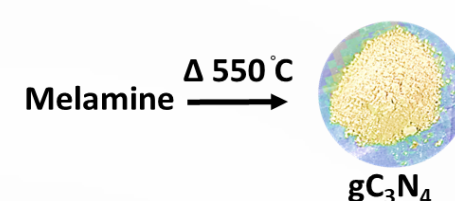
"Replacing synthetic polymer-based mulch with biomass-derived sheets"



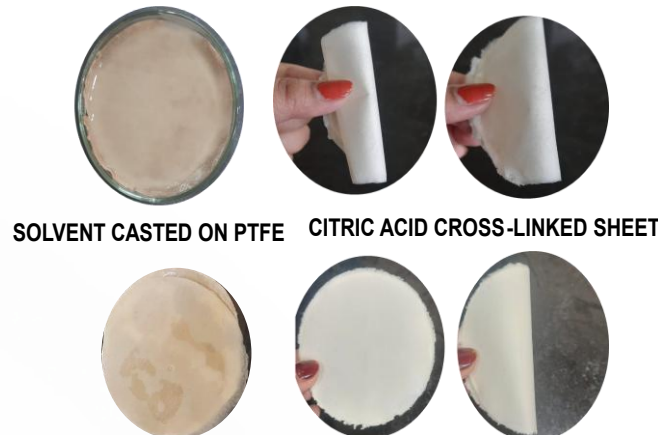
PHOTOCATALYST INTEGRATION

Choice of Photocatalyst: gC_3N_4

- Visible-light-responsive
- Metal-free and environmentally benign
- Narrow band gap (~2.8 eV)
- Good chemical stability

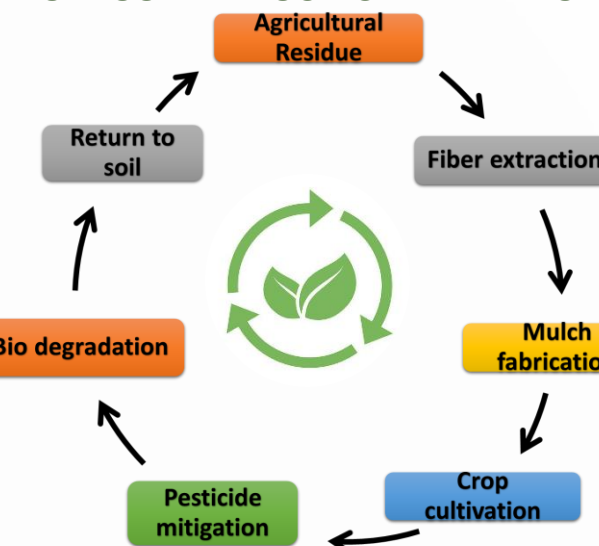


Synthesized via direct thermal polycondensation of Melamine



- ✓ Photocatalyst uniformly dispersed after homogenization
- ✓ Citric acid cross linking done to increase mechanical strength

07 CIRCULAR ECONOMY IMPACT

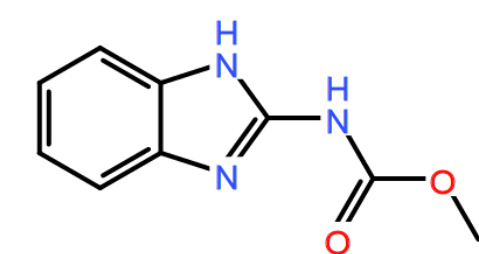
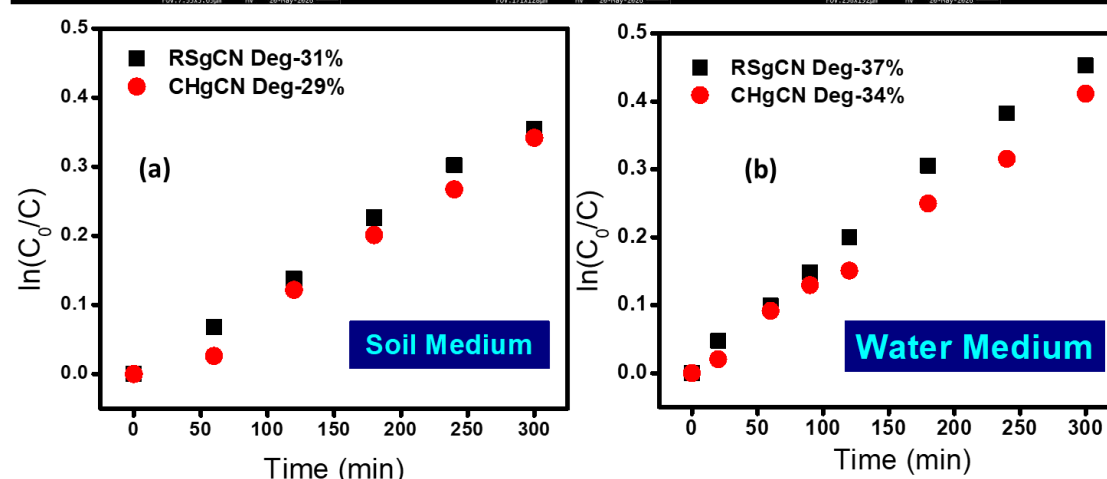
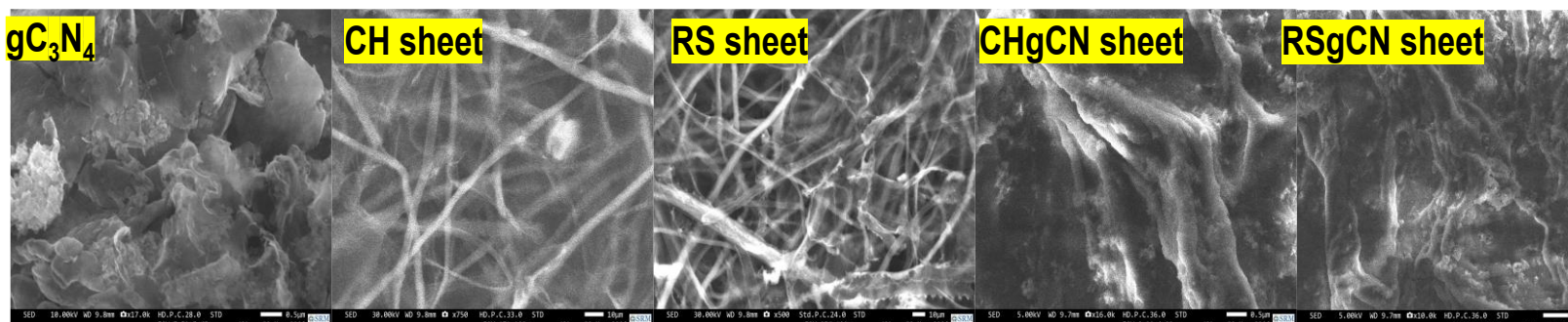
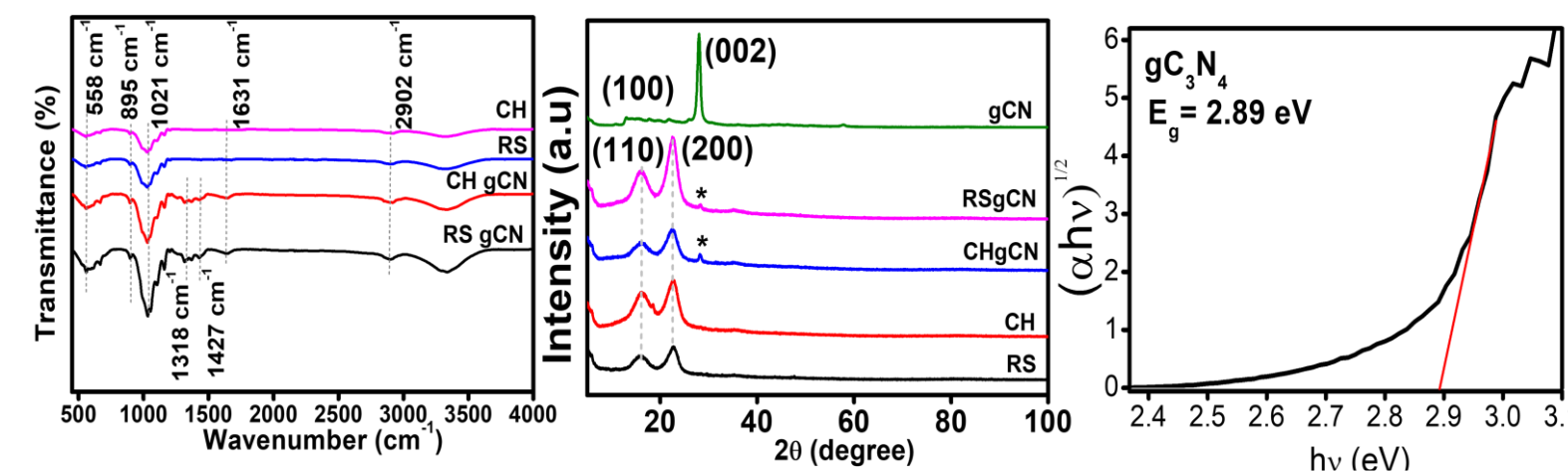


08 CONCLUSIONS

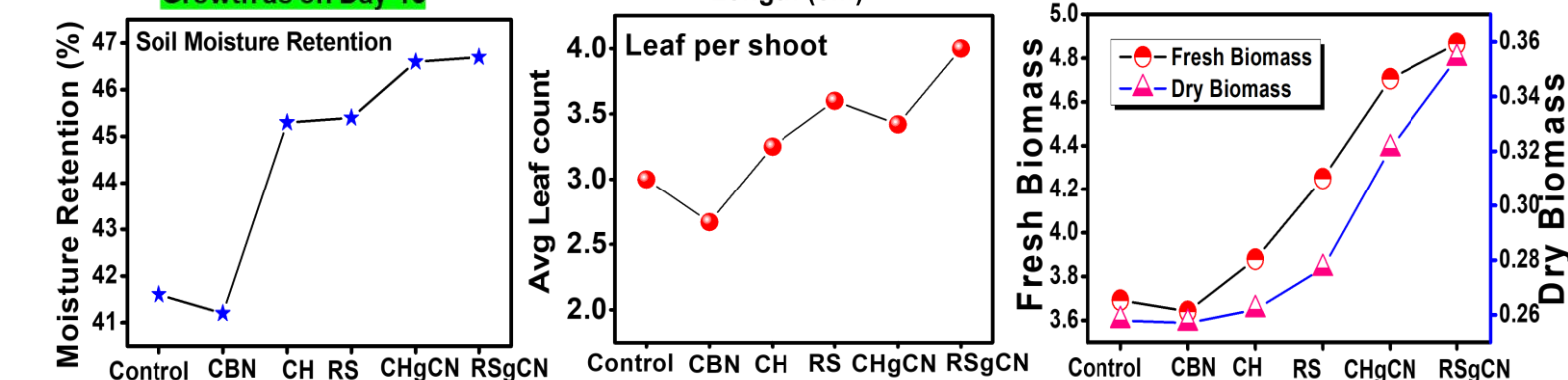
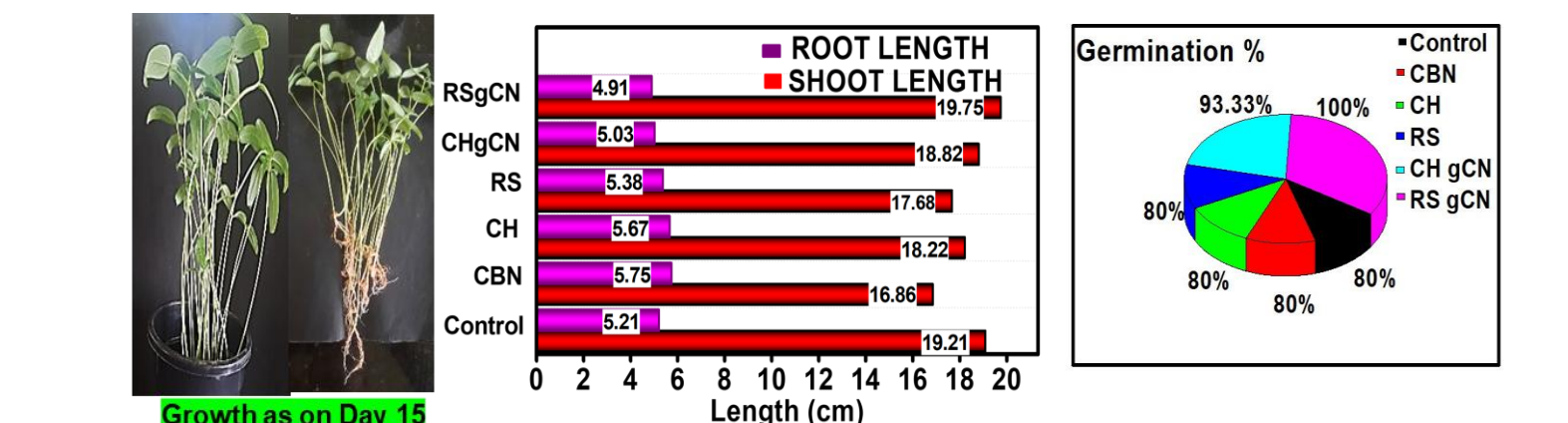
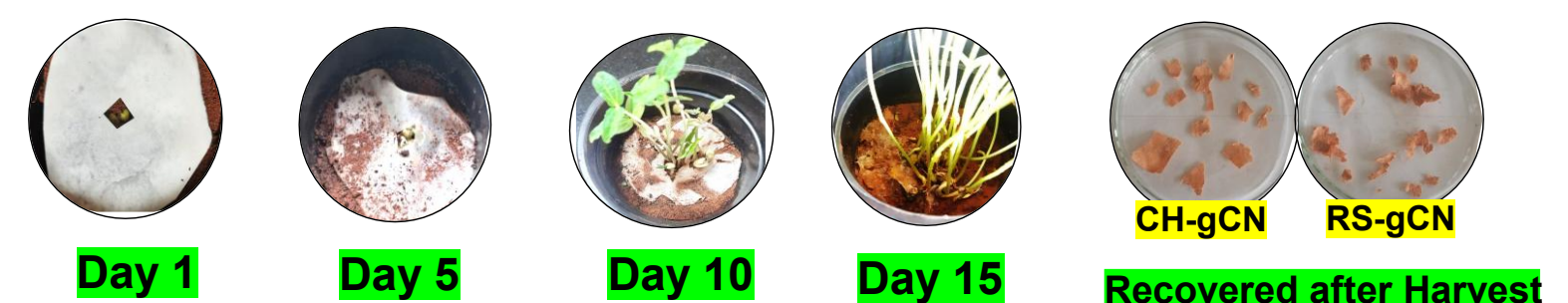
- Biomass-derived mulch sheets**
 - Agricultural waste valorisation
 - Cellulose-rich biodegradable matrix
 - Sustainable alternative to plastic mulch
- Visible-light photocatalytic platform**
 - gC_3N_4 integrated into biomass sheet
 - Retained soil moisture
 - Promoted plant growth
 - Active degradation of pesticide residues
- Agricultural and environmental relevance**
 - Reduced agro-plastic pollution
 - Potential mitigation of pesticide residue
 - Towards multifunctional smart mulching

Transforming agricultural mulch from a passive covering material into an active environmental remediation platform

05 FUNCTIONAL PERFORMANCE



06 PLANT RESPONSE (Pot cultivation studies with Vigna Radiata)



09 REFERENCES

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- Chen, Y. et al. Polylactic acid degradable mulching film modified by N-TiO₂/g-C₃N₄ photocatalyst for removal of carbendazim in water and soil under visible light. Journal of Environmental Management 380, 125135 (2025).
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10 ACKNOWLEDGEMENTS

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